

Semih Akin

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EDUCATION

Ph.D. Purdue University , Mechanical Engineering, (West Lafayette, USA)	2017 – 2022
M.S. Bursa Technical University , Mechanical Engineering, (Turkey)	2013 – 2016
B.S. Uludag University , Industrial Engineering, (Turkey) • <i>Double Major in Industrial Engineering</i>	2010 – 2013
B.S. Uludag University , Mechanical Engineering, (Turkey) • <i>Honor student, Ranked 1st in the class diploma</i>	2008 – 2013

RESEARCH & PROFESSIONAL EXPERIENCE

Assistant Professor , Rensselaer Polytechnic Institute, USA	Jan 2024 - Now
Post-Doctoral Associate , Purdue University, USA	2022 - 2023
Instructor- Ward A. Lambert Graduate Teaching Fellow - Purdue University, USA	2021 - 2022
Teaching Assistant , Purdue University, USA	2019 - 2021
Research Assistant , Purdue University, USA	2017 - 2021
Research Assistant , Bursa Technical University, Turkey	2013 - 2016

RESEARCH INTEREST

Additive Manufacturing: Cold spray additive manufacturing, Directed energy deposition, Multi-material 3-D printing, Aerosol jet printing

Surface Engineering: Surface-matter interaction, Meta-material surface deposition, Smart thin-films, Electroless deposition, Functional surface metallization of polymers

Manufacturing Intelligence: Physics-based and data-driven modeling, Sensor fusion, Digital twins

Space Manufacturing: Manufacturing for space and in-situ resource utilization

Cyber-Physical Manufacturing: Physical Unclonable Functions (PUFs), Spectral part authentication, Adaptive manufacturing, Predictive maintenance

Smart Structures and Energy Devices: Printed electronics, Smart textiles, Triboelectric nanogenerators, Piezoelectric nanogenerators, Lithium-ion batteries

HONORS, AWARDS & RECOGNITION

Research Awards:

- **ASME Reviewers of the Year**, *Journal of Micro and Nano Science and Engineering*, 2025
- **Outstanding Reviewer**, *Journal of Manufacturing Processes*, 2025
- **SigmaXI Scientific Research Honor Society** (Full Member), 2024
- **Outstanding Graduate Student Research Award**, Purdue University, CoE, 2023
- **International Research Awards** on Computer-Aided Design in Mechanical Engineering, 2023
- **Italian Packaging Technology Award** by the Italian Trade Agency, 2023
- **Graduate School Summer Research Grant**, Purdue University, CoE, 2022
- **Featured Article** in the Purdue News, (e-textiles for health monitoring), 2022
- **Master Thesis Scholarship** by the Technological Research Council of Turkey, 2015
- **Honor Student**, ranked 1st in Mechanical Engineering, Bursa Uludag University, 2013
- **Outstanding Student Scholarship** by the Turkish Automobile Factory (TOFAS), 2009-2013

Paper Awards:

- **Student Best Paper Award**, *International Workshop on Structural Health Monitoring*, 2025
- **Best Paper Award**, *Journal of Manufacturing Processes*, 2024.
- **Best Paper Award**, *International Mechanical Engineering Congress & Exposition*, (IMECE), 2024
- **Frontispiece Cover Article**, *Advanced Materials*, 2022
- **Editor's Choice Article**, *Journal of Thermal Spray Technology*, 2021
- **Best Paper Award**, *World Congress on Micro and Nano Manufacturing* (WCMNM), 2021

Teaching Awards:

- **Ward A. Lambert Graduate Teaching Fellowship**, Purdue University, 2022
- **Outstanding Graduate Teaching Award**, Purdue University Teaching Academy, 2022

Travel Awards:

- **National Science Foundation (NSF) travel award** for the WCMNM 2023
- **NSF Early-Career Travel Award** for NAMRC 51/MSEC 2023
- **NSF Student Travel Award** for the WCMNM 2019
- **Technical Trip Award** to Germany by the Durmazlar Machine Company, 2013

INTELLECTUAL PROPERTY (PATENTS)

8. K. Young, **S. Akin**, J. Samuel, "A method for controlled bonding and selective metallization in multi-polymer layered systems", (*U.S. Patent application*), (2026).
7. **S. Akin**, J. Jeon, "A Blockchain-coupled physical unclonable function (PUF) framework for robust part authentication, traceability, and cyber-physical security", (*U.S. Patent application*), (2025).
6. **S. Akin**, SH. Abir, J. Samuel, "Method for fabricating flexible metallized composite bacterial cellulose structures for energy harvesting and sensing applications", (*U.S. Patent application*), (2025).
5. MBG. Jun, J. Lee. H. Lee, **S. Akin**, C. Han, T. Gabor, Y. Sim, "Cold spray-enabled physically unclonable identifier and its spectral authentication via implicit neural representation", (*U.S. Patent application*), (2025).
4. **S. Akin**, J. Samuel, F. Kopsaftopoulos, J. Ren, P. Zhou, G. Saunders, "Method for enhanced adhesion across fully encapsulated metal-ceramic interfaces in additive manufacturing processes", (*U.S. Patent application*), (2025).
3. C.H. Lee, **S. Akin**, T. Chang, MBG. Jun, "Electronic textiles and systems and processes associated therewith", (*U.S. Patent application-pending*), (2023). [\[Link\]](#) 
2. MBG. Jun, **S. Akin**, "Cold spray printed flexible electronics and method for manufacturing the same" (*U.S. Patent application-pending*), (Active by 2044). [\[Link\]](#) 
1. C.H. Lee, T. Chang, **S. Akin**, MBG. Jun, L. Couetil, "Electronic textiles and methods for fabrication thereof", (*U.S. Patent*), (Active by 2043). [\[Link\]](#) 

EXTRAMURAL RESEARCH GRANTS (Akin's Share: \$536,942 → Total: \$1,524,133)

- **Funding Agency:** DURIP (Defense University Research Instrumentation Program)
Title: "A direct-writing platform for the development of smart structural systems with embedded sensing capabilities"
Role: Co-PI
Project budget = \$300,499 (Akin's share = 25%)
Project term: 01/01/26 - 01/01/27
- **Funding Agency:** New York Energy Storage Engine Projects
Title: "Dry-coating of lithium-ion battery anodes by cold spray"
Role: Lead PI
Project budget = \$223,634 (Akin's share = 50%)
Project term: 01/01/25 - 01/31/26
- **Funding Agency:** Defense Advanced Research Projects Agency (DARPA)
Title: "Convergent manufacturing of smart metal structures with embedded sensing capabilities"
Role: Co-PI
Project budget = \$1,000,000 (Akin's share = 35%)
Project term: 01/22/24 - 01/22/26

PUBLICATIONS SUBMITTED FOR PEER-REVIEW

†: Equal contribution

*: Corresponding author

1. S. Rahman, FT. Zohora, **S. Akin**^{*}, "Cold spray structural repair using lunar regolith-based feedstock".
2. K. Young, J. Dhar, J. Samuel[†], **S. Akin**^{*}, "Filament-fed additive friction stir deposition for polymer surface functionalization", **Journal of Manufacturing Process**, (Under revision).
3. J. Jeon, SH. Abir, S. Huang, J. Dhar, G. Richard, AR Ananda, F. Kopsaftopoulos, J. Samuel, **S. Akin**^{*}, "Multimaterial hybrid additive manufacturing of smart polymer structures with embedded electronics", **ACS Applied Electronic Materials**, (Under revision).
4. RM. Manoj, A. Anjan, J. Jeon, R. Sinha, V. Pongalat, R. Mukharjee, **S. Akin**, N. Koratkar^{*}, "Eliminating use of toxic N-methyl-2-pyrrolidone in battery processing", **ACS Applied Engineering Materials**, (Under revision).

PEER-REVIEWED JOURNAL PUBLICATIONS

†: Equal contribution

*: Corresponding author

36. FT. Zohora, J. Jeon, **S. Akin**^{*}, "Data-driven prediction of cold spray additive manufacturing via CFD-guided machine learning", **Results in Engineering**, (2026), (doi.org/10.1016/j.rineng.2026) [↗](#).
35. S. Rahman[†], SH. Abir[†], D. Fritz, J. Samuel, **S. Akin**^{*}, "Lunar regolith simulant-based triboelectric nanogenerators via solid-state additive manufacturing", **Applied Materials Today**, (2026), (doi.org/10.1016/j.apmt.2026.103417) [↗](#).
34. J. Ren, S. Huang, S. Rahman, P. Zhou, Y. Fan, B. Billanueva, G. Saunders, JT. Wen, S. Mishra, J. Samuel[†], F. Kopsaftopoulos^{*}, **S. Akin**^{*}, "A convergent manufacturing pathway for smart metallic structures with embedded sensing", **Progress in Additive Manufacturing**, (2026), (doi.org/10.1007/) [↗](#).
33. H. Lee, C. Han, T. Gabor, Y. Sim, **S. Akin**^{*}, MBG. Jun, Y. Jeon, J. Lee^{*}, "Cold spray-based secure and unique product identification with neural encoding: A full-stack framework for scalable authentication in manufacturing", **Journal of Intelligent Manufacturing**, (2026), ([10.1007/s10845-026-02864](https://doi.org/10.1007/s10845-026-02864)) [↗](#).
32. J. Jeon, **S. Akin**^{*}, "Physically unclonable surfaces enabled by cold spray deposition", **ACS Applied Materials & Interfaces**, (2026), (doi.org/10.1021/acsami.5c17570) [↗](#).
31. S. Rahman, **S. Akin**^{*}, J. Ren, P. Zhou, F. Kopsaftopoulos, J. Samuel, "Additively manufactured smart metallic structures with embedded sensing: A review", **Progress in Additive Manufacturing**, (2026), (doi.org/10.1007/s40964-025-01481-y) [↗](#).
30. S. Rahman, **S. Akin**^{*}, "Cold spray deposition of lunar regolith on polymeric substrates: A pathway toward in-situ resource utilization on the Moon", **Surfaces and Interfaces** (2026), (doi.org/10.1016/j.surfin.2025.108224) [↗](#).
29. C. Han, T. Gabor, H. Lee, J. Lee, **S. Akin**, MBG. Jun^{*}, "Pulsed cold spray system for physical unclon-

able function generation", **Procedia CIRP**, (2025), (doi.org/10.1016/j.procir.2025.02.269)

28. C. Han, T. Gabor, H. Lee, J. Lee, **S. Akin**, MBG. Jun*, "Pulsed cold spray system for physical unclonable function generation", **Procedia CIRP**, (2025), (doi.org/10.1016/j.procir.2025.02.269)

27. SH. Abir, C. Smith, J. Zorniter, J. Samuel*, **S. Akin***, "A composite bacterial cellulose for enhanced performance triboelectric and piezoelectric nanogenerators", **Nano Energy**, (2025), (doi.org/10.1016/j.nanoen.2025.111123)

26. T. Gabor, Y. Wang, **S. Akin**, F. Zhou, J. Chen, MBG. Jun*, "Design, modeling, and characterization of a pulsed cold spray", **Surface & Coatings Technology**, (2025), (doi.org/10.1016/j.surfcoat)

25. F. Zhou, S. Chen, **S. Akin**, T. Gabor, MBG. Jun*, "Real-time monitoring of thin film thickness and surface roughness using a single mode optical fiber", **Mechanical Systems and Signal Processing**, (2025), (doi.org/10.1016/j.ymssp.2024.112219)

24. J. Lee, **S. Akin***, Y. Kim, E. Kim, J. Nam, K. Song, MBG. Jun*, "A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing", **Manufacturing Letters**, (2024), (doi.org/10.1016/j.mfglet.2024.09.178)

23. **S. Akin**[†]*, T. Chang[†], S.H. Abir[†], Y. W. Kim, S. Xu, J. Lim, Y. Sim, J. Lee, J.T. Tsai, C. Nath, H. Lee, W. Wu, J. Samuel, C.H. Lee*, MBG. Jun*, "One-step fabrication of functionalized electrodes on 3D-printed polymers for triboelectric nanogenerators", **Nano Energy**, (2024), (doi.org/10.1016/)

22. DG. Ruzgar, **S. Akin**, S.Lee, J. Walsh, YH. Jeong, H.Lee, MBG. Jun*, "Highly flexible, conductive, and antibacterial surfaces toward multifunctional flexible electronics", **International Journal of Precision Engineering and Manufacturing Green Technology**, (2024), (doi.org/10.1007/s40684-024-00608-w)

21. **S. Akin***, S.Kim, C.K. Song, S.Y. Nam, MBG. Jun*, "Fully additively manufactured counter electrodes for dye-sensitized solar cells", **Micromachines**, (2024), (doi.org/10.3390/mi15040464)

20. JT. Tsai*, **S. Akin**, DF. Bahr, MBG. Jun, "A predictive modeling approach for cold spray metallization on polymers", **Surface & Coatings Technology**, (2024), (doi.org/10.1016/j.surfcoat.2024)

19. T. Gabor, **S. Akin**, MBG. Jun*, "Numerical studies on cold spray gas dynamics and powder flow in circular and rectangular nozzles", **Journal of Manufacturing Processes**, (2024), (*Best Paper Award*) (doi.org/10.1016/j.jmapro.2024.02.005)

18. Jeong H. Kim, **S. Akin**, MBG. Jun, Y. H. Jeong*, "Fabrication of electrospun nanofibers with spray direct-write conductive patterns", **Journal of the Korean Society for Precision Engineering**, (2024), (doi.org/10.7736/jkspe)

17. T. Chang[†], **S. Akin**[†], S. Cho[†], S. Lee, J. Lee, S. Lee, T. Park, S. Hong, T. Yu, Y. Ji, S. Gong, D.R. Kim, Y.L. Kim, MBG. Jun*, C.H. Lee*, "In-situ spray polymerization of conductive polymers for personalized e-textiles", **ACS Nano**, (2023), (doi.org/10.1021/acsnano.3c07283)

16. **S. Akin***, C. Nath, MBG. Jun, "Selective surface metallization of 3D-printed polymers by cold spray-assisted electroless deposition", **ACS Applied Electronic Materials**, (2023), (doi.org/10.1021/acsaelm.3c00893)

15. J. Lee, **S. Akin**, J. Walsh, H. Lee, MBG. Jun, Y. Shin*, “A Nitinol structure with functionally gradient pure titanium layers and hydroxyapatite over-coating for orthopedic implant applications”, **Progress in Additive Manufacturing**, (2023), (doi.org/10.1007/s40964) [🔗](#).
14. **S. Akin***, Y.W. Kim, S. Xu, C. Nath, W. Wu, MBG. Jun, “Cold spray direct writing of flexible electrodes for enhanced performance triboelectric nanogenerators”, **Journal of Manufacturing Processes**, (2023), (doi.org/10.1016/j.jmapro.2023.05.015) [🔗](#).
13. **S. Akin**, P. Wu, C. Nath, J. Chen, MBG. Jun*, “A study on converging-diverging nozzle design for supersonic spraying of liquid droplets towards nanocoating applications”, **ASME Journal of Manufacturing Science and Engineering**, (2023), (doi.org/10.1115/1.4062351) [🔗](#).
12. **S. Akin**, S. Jo, MBG. Jun*, “A cold spray-based novel manufacturing route for flexible electronics”, **Journal of Manufacturing Processes**, (2023), (doi.org/10.1016/j.jmapro.2022) [🔗](#).
11. **S. Akin**, S. Lee, S. Jo, DG. Ruzgar, JT. Tsai, MBG. Jun*, “Cold spray-based rapid and scalable production of printed flexible electronics”, **Additive Manufacturing**, (2022), (doi.org/10.1016/j.addma.2022.103244) [🔗](#).
10. Y.W. Kim*, **S. Akin**, H. Yun, S. Xu, W. Wu, MBG. Jun, “Enhanced performance of triboelectric nanogenerator and sensor via cold spray particle deposition”, **ACS Applied Materials & Interfaces**, (2022), ([doi/10.1021/acsami.2c09367](https://doi.org/10.1021/acsami.2c09367)) [🔗](#).
9. T. Gabor, H. Yun, **S. Akin**, K.H. Kim, J.K. Park, MBG. Jun*, “Continuous coaxial nozzle designs for improved powder focusing in direct laser metal deposition”, **Journal of Manufacturing Processes**, (2022), (doi.org/10.1016/j.jmapro.2022.08.03900) [🔗](#).
8. JT. Tsai, **S. Akin**, F. Zhou, MS Park, D.F. Bahr, MBG. Jun*, “Electrically conductive metallized polymers by cold spray and co-electroless deposition”, **ASME Open Journal of Engineering**, (2022), (doi.org/10.1115/1.4053781) [🔗](#).
7. T. Chang[†], **S. Akin**[†], M.K. Kim, L. Murray, S. Cho, L. Couetil, MBG. Jun*, C.H. Lee* “A Programmable dual regime spray for large-scale and custom-designed electronic textiles”, **Advanced Materials**, (2022), (doi.org/10.1002/adma.202108021) [🔗](#), (Cover Article), [Link] [🔗](#).
6. S. Jo, **S. Akin**, MS. Park, MBG. Jun*, “Selective metallization on glass surface by laser direct writing combined with supersonic particle deposition”, **Manufacturing Letters**, (2022), (doi.org/10.1016/j.mfglet.2021.07.009) [🔗](#).
5. **S. Akin**, P. Wu, JT. Tsai, C. Nath, J. Chen, MBG. Jun*, “A study on droplets dispersion and deposition characteristics under supersonic spray flow for nanomaterial coating applications”, **Surface & Coatings Technology**, (2021), (doi.org/10.1016/j.surfcoat.2021.127788) [🔗](#).
4. JT. Tsai, **S. Akin**, F. Zhou, DF. Bahr*, MBG. Jun, “Establishing a cold spray particle deposition window on polymer substrate”, **Journal of Thermal Spray Technology**, (2021), (doi.org/10.1007/s11666-021-01179-x) [🔗](#), (Editor’s choice article)
3. **S. Akin**, JT. Tsai, MS. Park, YH. Jeong, MBG. Jun*, “Fabrication of electrically conductive patterns on ABS polymer using low-pressure cold spray and electroless plating”, **ASME Journal of Micro and Nano-Manufacturing**, (2020), (doi.org/10.1115/1.4049578) [🔗](#).
2. **S. Akin**, T. Gabor, S. Jo, H. Joe, JT. Tsai, Y. Park, C.H. Lee, MS. Park, MBG. Jun*, “Dual regime spray

deposition based laser direct writing of metal patterns on polymer substrates”, **ASME Journal of Micro and Nano-Manufacturing**, (2020), (doi.org/10.1115/1.4046282)

1. **S. Akin***, Y. Kara, “An assessment of wind power potential along the coast of Bursa, Turkey: A wind power plant feasibility study for Gemlik Region”, **Journal of Clean Energy Technologies**, (2017), ([doi:10.18178/jocet.2017.5.2.352](https://doi.org/10.18178/jocet.2017.5.2.352)) .

CONFERENCE PROCEEDINGS & PRESENTATIONS

Ψ: Presenter

*: Corresponding author

30. FT. Zohora^Ψ, **S. Akin***, "Physics-informed neural network for particle temperature prediction in cold spray", *ASME International Manufacturing Science and Engineering Conference (MSEC)* , Oregon, USA **2027**, (Under Review).

29. S. Rahman^Ψ, FT. Zohora, **S. Akin***, " Lunar regolith impact, fragmentation, and deposition in cold spray additive manufacturing: A smoothed particle hydrodynamics study", *ASME International Manufacturing Science and Engineering Conference (MSEC)* , Oregon, USA **2027**, (Under Review).

28. J. Jeon, **S. Akin***,^Ψ, " Cold spray-enabled physical fingerprints for anti-counterfeiting", *The Minerals, Metals & Materials Society (TMS)* , Orlando, Florida, USA, (**2027**), (Accepted).

27. S. Rahman, J. Phillips, ND. Walker, D. Cepeloglu, M. Koca, O. Tumuklu*, **S. Akin***, “Structural repair of aluminum 6061 alloy using cold spray additive manufacturing”, *ASME International Mechanical Engineering Congress & Exposition, IMECE* , British Columbia, Canada, (**2026**), (Accepted).

26. S. Rahman^Ψ, **S. Akin***, “Cold spraying of lunar regolith composite powders toward in-space electronics manufacturing”, *ASME International Manufacturing Science and Engineering Conference (MSEC)* , Penn State, Pennsylvania, USA, **2026**, (In Press).

25. J. Jeon ^Ψ, FT. Zohora, **S. Akin***, "Direct-writing of physical unclonable function (PUF)-augmented QR-codes for cyber-physical authentication", *North American Manufacturing Research Conference, NAMRC-53* , Penn State, Pennsylvania, USA, (**2026**), (doi.org/10.1016/j.mfglet.2026.08.030)

24. J. Ren, S. Huang, J. Samuel, F. Kopsaftopoulos, **S. Akin***,^Ψ, "Convergent manufacturing of smart metallic structures", (**2026**), *Symposium on Convergent Manufacturing at Point of Need* .

23. J. Jeon, A. Wong, M. Koca, F. Thompson, O. Tumuklu, **S. Akin***,^Ψ, "Conductivity-tunable reactive aerosol jet metallization of textiles", *ASME International Mechanical Engineering Congress & Exposition, IMECE* , Memphis, TN, USA, (**2025**), ([10.1115/IMECE2025-164431](https://doi.org/10.1115/IMECE2025-164431)) .

22. S. Huang^Ψ, J. Ren, P. Zhou, S. Rahman, K. Young, S.S. Rahman, S. Mishra, J. Samuel, F. Kopsaftopoulos, **S. Akin***, "A multifunctional smart metal beam with sub-surface embedded sensors for real-time structural health monitoring", *International Workshop on Structural Health Monitoring (IWSHM)* , Stanford University, CA, USA, (**2025**), ([10.12783/shm2025/37491](https://doi.org/10.12783/shm2025/37491)) .

21. P. Zhou, J. Ren^Ψ, J.S. Schure, S. Huang, S. Rahman, J. Samuel*, **S. Akin***, F. Kopsaftopoulos*, "Embedded piezoelectric sensing for metallic components: A novel SHM architecture for self-aware structures", *International Workshop on Structural Health Monitoring (IWSHM)* , Stanford University, CA, USA, (**2025**), ([10.12783/shm2025/37301](https://doi.org/10.12783/shm2025/37301)) , (*Best Paper Award*).

20. C. Han, T. Gabor, H. Lee, J. Lee, **S. Akin**, MBG. Jun^{*,Ψ}, "Pulsed cold spray system for physical unclonable function generation", *CIRP Conference on Electro Physical and Chemical Machining (ISEM XXII)* [\[↗\]](#), University of British Columbia, Vancouver, BC, Canada, (2025).
19. S. Chen^Ψ, F. Zhou, BN. Reggetz, EG. Lee, MA. Virji, AA. Afshari, MBG. Jun, **S. Akin**^{*}, "Polymer metallization via cold spray: an investigation into the effects of particle hardness and morphology", *North American Manufacturing Research Conference, NAMRC-53* [\[↗\]](#), Greenville, South Carolina, USA, (2025), (doi.org/10.1016/j.mfglet.2025.06.039) [\[↗\]](#)
18. M. Muhtadin, JT. Tsai^{*,Ψ}, **S. Akin**, "Additive manufacturing of radially oriented gyroid carbon fiber composites for low-temperature thermal absorber applications", *North American Manufacturing Research Conference, NAMRC-53* [\[↗\]](#), Greenville, South Carolina, (2025), (doi.org/10.1016/j.mfglet.2025.06.096) [\[↗\]](#)
17. J. Lee, **S. Akin**, J. Walsh, H. Lee, MBG. Jun, Y. Shin^{Ψ*}, "Functionally gradient nitinol structure with pure titanium layers and hydroxyapatite over-coating for orthopedic implant applications", *TMS Annual Meeting & Exhibition* [\[↗\]](#), Las Vegas, Nevada USA, (2025).
16. Y.W. Kim, **S. Akin**^{Ψ*}, MBG. Jun, J. Sutherland, "Cold spray-produced functional surfaces for triboelectric nanogenerators", *ASME International Mechanical Engineering Congress & Exposition, IMECE* [\[↗\]](#), Portland, OR, USA, (2024), (doi.org/10.1115/IMECE2024-145320) [\[↗\]](#), (Best Paper Award).
15. S. Jo, **S. Akin**, M.S. Park, M.B.G. Jun^{*,Ψ}, "A study on supersonic spray-assisted laser-induced ultra-fine selective metallization of glass surface," *World Congress on Micro and Nano Manufacturing (WCMNM)* [\[↗\]](#), Pattaya, Thailand, 2024.
14. S. Jo, **S. Akin**, H. Yun, M. Park, MBG. Jun^{Ψ*}, "Laser-assisted ultrafine selective metallization of glass surface using supersonic spray deposition", *International Conference on Precision Engineering and Sustainable Manufacturing* [\[↗\]](#), Chiang Mai, Thailand, (2025).
13. J. Lee^Ψ, **S. Akin**^{*}, Y. Kim, E. Kim, J. Nam, K. Song, MBG. Jun^{*}, "A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing", *North American Manufacturing Research Conference, NAMRC-52* [\[↗\]](#), Knoxville, Tennessee, USA, (2024) (doi.org/10.1016/j.mfglet.2024.09.178) [\[↗\]](#)
12. JT. Tsai^{Ψ*}, **S. Akin**, DF. Bahr, MBG. Jun, "A predictive modeling for cold spray deposition and the resulting microstructure toward additive manufacturing using polymeric templates", *International Thin Films Conference (TACT-2023)* [\[↗\]](#), Taipei, Taiwan, (2023).
11. **S. Akin**^{Ψ*}, MBG. Jun, "Additively manufactured counter electrodes for dye-sensitized solar cells", *World Congress on Micro and Nano Manufacturing (WCMNM)* [\[↗\]](#), Evanston, IL, USA (2023).
10. MBG. Jun^{Ψ*}, **S. Akin**, "Unleashing the potential of cold spray additive manufacturing in triboelectric energy harvesting", *US-Korea Conference on Science, Technology and Entrepreneurship* [\[↗\]](#).
9. **S. Akin**^{Ψ*}, Y.W. Kim, S. Xu, C. Nath, W. Wu, MBG. Jun, "Cold spray direct writing of flexible electrodes for enhanced performance triboelectric nanogenerators", *North American Manufacturing Research Conference, NAMRC* [\[↗\]](#), New Brunswick, New Jersey, USA, (2023).
8. **S. Akin**^{Ψ*}, P. Wu, C. Nath, J. Chen, MBG. Jun^{*}, "A study on the effect of nozzle geometrical parameters on supersonic cold spraying of droplets", *ASME International Manufacturing Science and Engineering Conference*, (2022), West Lafayette, Indiana, USA, (doi.org/10.1115/MSEC2022-85703) [\[↗\]](#).

7. T. Gabor^Ψ, **S. Akin**, JT. Tsai, S. Jo, F. Najjar, MBG. Jun*, “Numerical studies on cold spray particle deposition using a rectangular nozzle”, *ASME MSEC*, (2022), *West Lafayette, Indiana, USA*, (doi.org/10.1115/MSEC2022-85673) .
6. **S. Akin**^{Ψ*}, J.H. Kim, MBG. Jun*, “Electrically conductive textiles based on decoupled atomized spray coating and electroless plating”, *International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM)* , *South Korea*, (2021).
5. S. Jo, **S. Akin**, MS. Park, MBG. Jun^{Ψ*}, “An integrated method for selective metallization on glass surface: Laser direct writing coupled with supersonic spray coating”, *World Congress on Micro and Nano Manufacturing (WCMNM)* , *IIT Bombay, India*, (2021), (*Best Paper Award*).
4. T. Gabor^Ψ, H. Joe, **S. Akin**, KH. Kim, JK. Park, MBG. Jun*, “Numerical investigations of various coaxial nozzle designs for direct laser deposition”, *ASME International Manufacturing Science and Engineering Conference (MSEC)*, *Cincinnati, Ohio, USA*, (2020), (doi.org/10.1115/MSEC2020-8444) .
3. JT. Tsai^Ψ, **S. Akin**, F. Zhou, DF. Bahr, MBG. Jun*, “Simulation and characterization of cold spray deposition of metal powders on polymer substrate electrically conductive application”, *ASME International Manufacturing Science and Engineering Conference*, *Cincinnati, Ohio, USA*, (2020), (doi.org/10.1115/MSEC2020-8461) .
2. **S. Akin**^Ψ, JT. Tsai, MS. Park, YH. Jeong, MBG. Jun*, “Fabrication of electrically conductive patterns on ABS polymer using low-pressure cold spray and electroless plating”, *ASME International Manufacturing Science and Engineering Conference*, *Cincinnati, Ohio, USA*, (2020), (doi.org/10.1115/) .
1. **S. Akin**^Ψ, T. Gabor, S. Jo, H. Joe, JT. Tsai, Y. Park, CH. Lee, MS. Park, MBG. Jun*, “Dual regime spray deposition based laser direct writing of metal patterns on polymer substrates”, *The 3rd World Congress on Micro and Nano-Manufacturing*, *Raleigh, North Caroline, USA*, (2019), (*WCMNM-2019*) .

POSTER PRESENTATIONS ---

6. SH. Abir, J. Samuel, **S. Akin**, “Metal-embedded bacterial cellulose for triboelectric energy harvesting”, *ASME Manufacturing Science and Engineering Conference (MSEC)* , *Greenville, USA*, (2025).
5. **S. Akin**, DA. Borca-Tasciuc, W. Ji, F. Kopsaftopoulos, A. Maniatty, K. Panneerselvam, C. Picu, A. Svirskey, “Curriculum integration through collaborative teaching”, *ASME International Mechanical Engineering Congress & Exposition, IMECE* , *Portland, OR, USA*, 2024, (*Best Poster Award*).
4. B. Reggetz, A. Virji, MBG. Jun, **S. Akin**, D. Hard, EG. Lee, “Assessment of cold spray powder emissions in a controlled laboratory Setting”, *Cold Spray Action Team (CSAT)* , *Worcester, MA, USA* (2024).
3. B. Reggetz, EG. Lee, A. Virji, S. Friend, MBG. Jun, **S. Akin**, “Cold spray powder emissions in a laboratory setting”, *AIHA Connect* , (2024).
2. T. Chang, **S. Akin**, L. Couetil, MBG. Jun, C.H. Lee “Dual regime spray of functional nanomaterials for electronic textiles”, *Material Research Society (MRS)* , *Hawaii, HI, USA* (2022), (*Best Poster Award*).
1. **S. Akin**, JT. Tsai, H. Joe, H. Joe, MBG. Jun, “Smart thin film on polymer and textile substrates by controlled spray and electroless plating”, *NextFlex*, *IN, USA*, (2020).

TEACHING & MENTORING EXPERIENCE

Instructor:

Rensselaer Polytechnic Institute

- **MANE 6962:** Additive Manufacturing *Spring 2026*
Instructor Rating: 4.78/5 Course Rating 4.72/5
- **ENGR 4710/MANE 4610:** Manufacturing Processes and Systems I *Fall 2025*
- **ENGR 4720/MANE 4620:** Manufacturing Processes and Systems II *Spring 2025*
Instructor Rating: 5/5
- **ENGR 2050:** Introduction to Engineering Design *Spring 2024, Fall 2024*
Instructor Rating: 4.3/5

Purdue University, West Lafayette

Aug 2022-Dec 2022

Instructor as the *Ward A. Lambert Fellow*

- **ME 354:** Machine Design
Instructor Rating: 4.4/5

Teaching Assistant:

Purdue University, West Lafayette

2019-2022

- **ME 352:** Machine Design I (Fall 2019, Spring 2020, Spring 2022)
- **ME 354:** Machine Design II (Fall 2020, Spring 2021, Fall 2021)

Bursa Technical University, TURKEY

2013- 2016

- Computer-aided design (CAD), Thermodynamics, Machine Laboratory, Senior Design Project

INDUSTRIAL EXPERIENCE

Intern at the **OYAK-RENAULT Automotive Company** , TURKEY

2012-2013

- Assisted a project from concept to minimize quality errors in vehicle batteries.
- Collected and analyzed data on quality errors of the vehicle batteries.
- Designed the software for quality control of the batteries.

TECHNICAL SKILLS

Programming languages: Python, MATLAB

Engineering software:

- **Computer-aided design (CAD):** Solidworks, CATIA, NX, AutoCAD, SpaceClaim
- **Computer-aided engineering (CAE):** ANSYS (Workbench, Fluent), Abaqus, HyperMesh
- **Other:** MS Office, L^AT_EX, Overleaf, Jupyter Notebook, Google Colab, OriginPro, MS Visio

INVITED TALKS & SEMINARS

1. **"Solid-state additive manufacturing: Building without melting"**
Union College, Mechanical Engineering Department, Schenectady, NY, USA (June 2025)
2. **"Additive Manufacturing of Functional Smart Surfaces"**
RPI MANE Department Graduate Seminar (2024)
3. **"Spray-Based Additive Manufacturing of Functional Smart Surfaces"**
University of Illinois Chicago (UIC), Department of Mechanical and Industrial Engineering, October (2023)
4. **"Additively Manufactured Counter Electrodes for Dye-Sensitized Solar Cells"**
World Congress on Micro and Nano Manufacturing (WCMNM), Evanston, IL, USA (2023)
5. **"Cold Spray Direct Writing of Flexible Electrodes for Enhanced Performance TENGs"**
North American Manufacturing Research Conference (NAMRC), New Jersey, USA (2023)
6. **"A Study on the Effect of Nozzle Geometrical Parameters on Supersonic Cold Spraying of Droplets"**
ASME International Manufacturing Science and Engineering Conference (MSEC), West Lafayette, Indiana, USA (2022)
7. **"Electrically Conductive Textiles Based on Decoupled Spray Coating and Electroless Plating"**
International Symposium on Precision Engineering and Sustainable Manufacturing (PRESM), South Korea (2021)
8. **"Fabrication of Electrically Conductive Patterns on ABS Polymer Using Low-Pressure Cold Spray and Electroless Plating"**
ASME International Manufacturing Science and Engineering Conference, Cincinnati, Ohio, USA (2020)
9. **"Dual Regime Spray Deposition-Based Laser Direct Writing of Metal Patterns on Polymer Substrates"**
World Congress on Micro and Nano-Manufacturing, Raleigh, North Carolina, USA (2019)

PROFESSIONAL SOCIETY MEMBERSHIP

- American Society of Mechanical Engineering (ASME)
- Society of Manufacturing Engineers (SME)
- SigmaXI Scientific Research Honor Society (Full Member)

PROFESSIONAL SERVICES

Journal Paper Peer-Reviewer:

- | | |
|--|---|
| • Journal of Manufacturing Process | • Nature Communications Materials |
| • ASME Journal of Micro and Nano Science and Engineering | • Energy Technology |
| • Journal of Manufacturing and Materials Processing | • Applied Mechanics |
| • Surface & Coatings Technology | • International Journal of Heat and Mass Transfer |
| • Applied Surface Science Advances | • Scientific Reports - Nature |
| • Applied Surface Science | • Materials Letters |
| • Surfaces & Interfaces | • Micromachines |
| • Structural Health Monitoring | • Electronics |
| • Additive Manufacturing | • Processes |
| | • Coatings |

Conference Reviewer:

- ASME International Mechanical Engineering Conference and Exposition (IMECE, 2026)
- ASME Manufacturing Science and Engineering Conference (MSEC, 2026)
- Additive Manufacturing Conference (AMC, Turkey), 2026
- ASME International Mechanical Engineering Conference and Exposition (IMECE, 2025)
- ASME Manufacturing Science and Engineering Conference (MSEC, 2025)
- World Congress on Micro-and Nano-Manufacturing (WCMNM-2025)
- North American Manufacturing Research Conference (NAMRC-53, 2025)
- North American Manufacturing Research Conference (NAMRC-52, 2024)
- North American Manufacturing Research Conference (NAMRC-51, 2023)

- World Congress on Micro-and Nano-Manufacturing (WCMNM-2023)

Services:

- Vice Chair, “*Manufacturing Equipment and Automation Technical Committee*”, ASME MED, 2026 - Present.
- Symposium Organizer, “*Advancing In-Space Manufacturing: Capabilities, Infrastructure, and Applications*”, ASME MSEC 2027
- Symposium Organizer, “*Recent Advancements in Solid-State Materials & Manufacturing Processes*”, ASME IMECE 2026
- Symposium Organizer, “*Advances in Solid-state Additive Manufacturing*”, ASME/MSEC 2026
- Judge, Student Poster Competition, ASME MSEC 2026.
- Session Chair, “*Smart Manufacturing and Cyberphysical Systems*”, SME NAMRC-2026
- Session Chair, “*Additive Manufacturing*”, MSEC 2026.
- Symposium Organizer, “*Advances in Manufacturing of Thin Films and Coatings*”, ASME/MSEC 2025
- Session Chair, “*Additive Manufacturing*”, SME NAMRC-2025
- Judge, ASME/SME Student Manufacturing Design Competition, 2025
- Head volunteer, ASME MSEC/SME NAMRC-2022

GRADUATE STUDENT MENTORING

- **Advisor - Ph.D. Students:**

Sazedur Rahman (Aug 2024 - Present), Jaehun Jeon (Spring 2025 - Present), Fatema-Tuj- Zohora (Fall 2025 - Present), Tajwar Razib (Fall 2026 - Present)

- **Advisor - Master of Eng. Students:**

Akshay Rao Ananda (Fall 2026 - Present), Kate Goldstein (Spring 2026 - Present), Leo Woytomich (Spring 2026 - Present), Nicole Dipoppo (Fall 2026 - Fall 2027), Jules Philips (Fall 2025 - Spring 2026), Nicholas Walker (Fall 2025 - Spring 2026), Faydia Thompson (Spring 2025 - Fall 2025)

- **Mentor - Ph.D. Students:**

Jinhan Ren, Joni C. Dhar, Shamim H. Abir (Jan 2024 - Present)

- **Mentor - Master of Science Students:**

Kyle Young (Fall 2024 - Spring 26)

- **Mentor - Master of Eng. Students:**

Charli Smith, Jared Zornitger (Spring 2024 - Fall 2024)

UNDERGRADUATE STUDENT MENTORING

- **Advisor - Undergraduate Research Students:**

Jiwoo Jung (Fall 2026 - Present)

Alex Wong (Fall 2024 - Present)

Grace Richard, Akshay Rao Ananda (Fall 2025 - Spring 2026)

Brandon Villanueva, Zach Goncalves, Zhi Guan, Travis Johnson (Spring 2024)

Hongfei Liu, Hongru Liu, (Fall 2024 - Spring 2025)

Ph.D. COMMITTEE MEMBER

Shinan Huang	RPI	2026 - Present
Jinhan Ren	RPI	2026 - Present
Rohit Monaj	RPI	2025 - Present
Sk. Shamim H. Abir	RPI	2025 - Present
Joni C. Dhar	RPI	2024 - Present
Chieloka Ibekwe	RPI	2024 - 2026
Sifat Ullal	RPI	2024 - 2026
Max Pranompont	RPI	2024 - 2026
Apurva Anjan	RPI	2024 - 2026

SELECTED MEDIA COVERAGE

- "ASME Reviewers of the Year", 2025. [\[Link1\]](#) [\[Link2\]](#)
- "SME Journal Awards: Best Paper Award, Journal of Manufacturing Processes", 2025. [\[Link\]](#)
- "Journal Awards: Outstanding Reviewer, Journal of Manufacturing Processes", 2025. [\[Link\]](#)
- "Titomic Breakthrough in Lithium-ion Electrode Manufacturing", 2025. [\[Link\]](#)
- "Rensselaer Polytechnic Institute & Titomic working to develop cold spray additive solution for lithium-ion battery electrode manufacturing", 2025. [\[Link\]](#)
- "National Manufacturing Day at Rensselaer Polytechnic Institute", 2025. [\[Link\]](#)
- "Outstanding Research Award *Purdue University, College of Engineering*, 2023. [\[Link\]](#)
- "Remote horse slicker monitors chronic health conditions" *Veterinary33*, July 2022. [\[Link\]](#)
- "How do you test for equine asthma and heart disease using a remote horse slicker? Put the horse on a treadmill" *Purdue Research News*, April 2022. [\[Link\]](#)
- "Specially designed slicker captures horse's vital signs on a laptop via Bluetooth" *Phys.org*, February 2022. [\[Link\]](#)
- "Horse slicker may help tell of animal's chronic diseases" *Newsbug*, February 2022. [\[Link\]](#)